Inverted MAML Card 1: Round 13 Thermal Bit-Slicing Report

Hardware: Card 1 (T120+T20, 1k/2.2k) | **FW:** v2.2-MAML (0x20261030) | **Params:** 40% PWM, K=1, LR=0.055, L2=1920, Mom=0.85, Clamp=±82 LSB

1. Round 13 Summary & Cumulative Comparison (R1–R12)

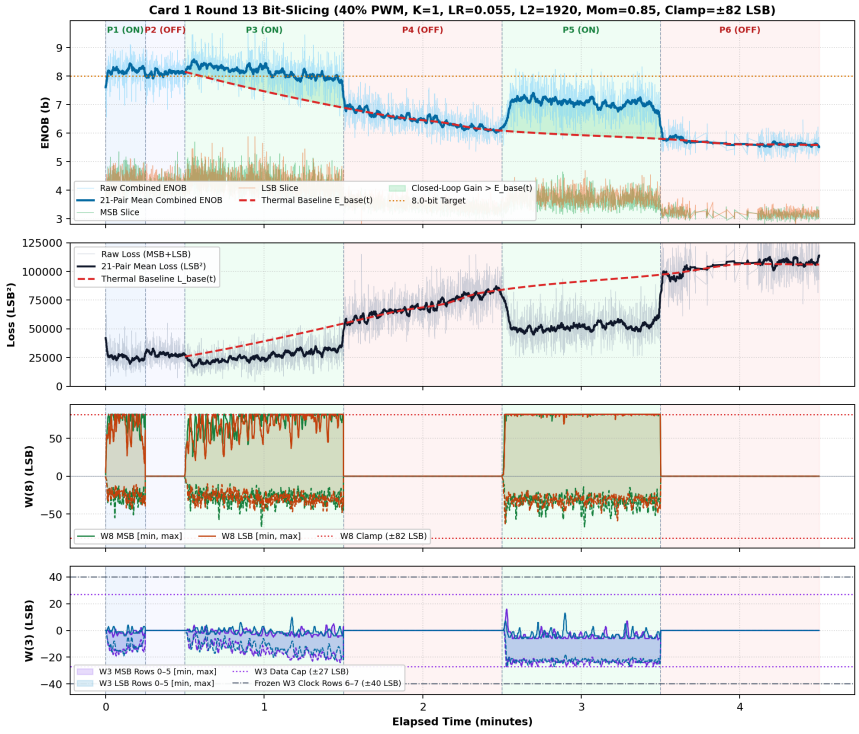
• **Cold Verification:** Phase 1 ON = **9.45 b**, Phase 2 OFF = **9.45 b** (W8=0 strictly).

• **Round 13 Detrended Trend Metrics:** Lift vs E_base = **+1.04 b**, Loss Ratio = **1.75x**, Climb = **+0.83 b**, Jitter = **0.42 b**, Recovery = **0.00**, Score = **+1.721**.

Rnd	LR	L2	Mom	Clamp	Lift	LossR	Climb	Jitt	Recov	Score
R5dc_fix	0.055	1792	0.85	±85	+1.12b	1.77x	+1.37b	0.42b	0.00	+2.393
R6dc_fix	0.055	1792	0.8	±90	+0.99b	1.72x	+1.02b	0.40b	0.00	+1.770
R7dc_fix*	0.055	2048	0.85	±75	+1.04b	1.74x	+1.17b	0.40b	0.50	+2.665
R8dc_fix	0.055	2048	0.85	±75	+1.16b	1.85x	+0.68b	0.42b	0.00	+1.855
R9dc_fix	0.055	1920	0.85	±75	+0.81b	1.59x	+0.66b	0.38b	0.50	+1.756
R10dc_fix	0.055	1920	0.85	±78	+0.96b	1.67x	+1.13b	0.40b	0.00	+1.898
R11dc_fix	0.055	1920	0.85	±80	+0.95b	1.70x	+0.98b	0.42b	0.00	+1.810
R12dc_fix	0.055	1920	0.85	±82	+1.04b	1.75x	+0.83b	0.42b	0.00	+1.721

2. Phase-by-Phase Telemetry (3,856 Pairs)

Phase	Regime	N	Start->Late (Mean)	Δ vs E_base	Loss Start->Late	W8 Pk [M,L]	W3 Pk	W8 nz
P1	Cold / Regler ON	321	7.84->8.24 (8.19b)	-	34257->23453	[82,82]	19	321
P2	Cold / Regler OFF	290	7.92->8.13 (8.10b)	-	32342->27128	[0,0]	0	0
P3	Warm / Regler ON	1038	8.27->7.97 (8.16b)	+0.88 b	24313->30703	[82,82]	24	1038
P4	Warm / Regler OFF	902	6.79->6.20 (6.47b)	+0.00 b	58542->78810	[0,0]	0	0
P5	Warm / Regler ON	820	6.29->6.99 (7.04b)	+1.14 b	74366->53692	[82,82]	27	820
P6	Warm / Regler OFF	485	5.77->5.59 (5.66b)	-0.00 b	99187->107800	[0,0]	0	0



3. All W(8) (7x6 MSB+LSB) & W(3) (8x6 MSB+LSB) Weight Dynamics — Round 13

• **Full 180-Synapse Per-Pair Capture:** Every individual weight in **Matrix 8 MSB (42 weights)**, **Matrix 8 LSB (42 weights)**, **Matrix 3 MSB (36 active + 12 clock)**, and **Matrix 3 LSB (36 active + 12 clock)** is plotted below across all columns `Col 0..5`. W(8) enforces anti-windup clamping at ±82 LSB, while W(3) rows 0–5 stay within ±27 LSB so rows 6–7 clock staircases remain locked at ±40 LSB.

Round 13 Full Hardware Weight Telemetry: ALL 84 W(8) [7x6 MSB+LSB] & ALL 96 W(3) [8x6 MSB+LSB] Synapses

